

3. FUNCTIONS

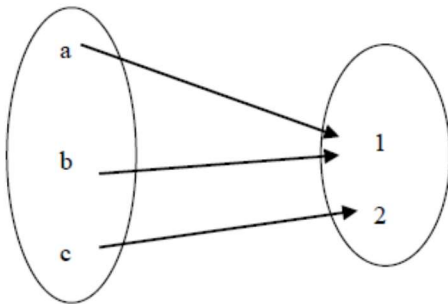
Functions play a central role in Mathematics where they are used to describe any process whereby the elements of one set are transformed into the elements of another.

Functions

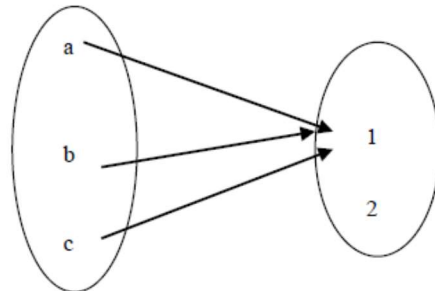
A **function** from a set A to set B is a binary relation in which every element of A is associated with a uniquely specified element of B . In other words, for each $a \in A$, there is precisely one pair of the form (a, b) .

In terms of a digraph, a function is a relation such that there is precisely one arrow leaving every element of A .

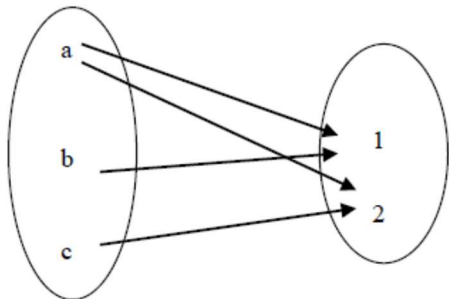
For example:



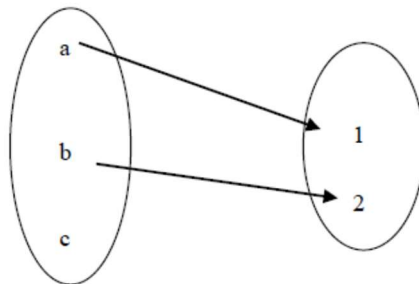
is a function



is a function



is not a function



is not a function

Example 1:

Determine whether each of the following relations between the sets $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$ represents a function.

- (a) $R = \{(a, 1), (a, 2), (b, 3), (c, 2)\}$
- (b) $S = \{(a, 1), (b, 2), (c, 1)\}$
- (c) $T = \{(a, 1), (c, 2)\}$

Solution:

- (a) R is not a function since two different elements of B (1 and 2) are associated with the element a .
- (b) S is a function since every element of A is associated with a uniquely specified element of B .
- (c) T is not a function, since no element of B is associated with b .

Example 2:

Determine if each of the following relations is a function.

- (a) $R = \{(x, y): x \text{ is a brother of } y\}$ on the set P of all people.
- (b) $S = \{(x, x^2): x \in \mathbb{Z}\}$
- (c) $T = \{(x, y): x = y^2, x, y \in \mathbb{R}\}$

Solution:

- (a) A given person x may have several brothers and sisters or none at all, hence R is not a function on P .
- (b) For any given integer x , the value x^2 is uniquely determined, hence S is a function.
- (c) Note: $(4, 2) \in T$ and $(4, -2) \in T$ hence, T is not a function. Also note that there are no pairs of the form (x, y) when $x < 0$.

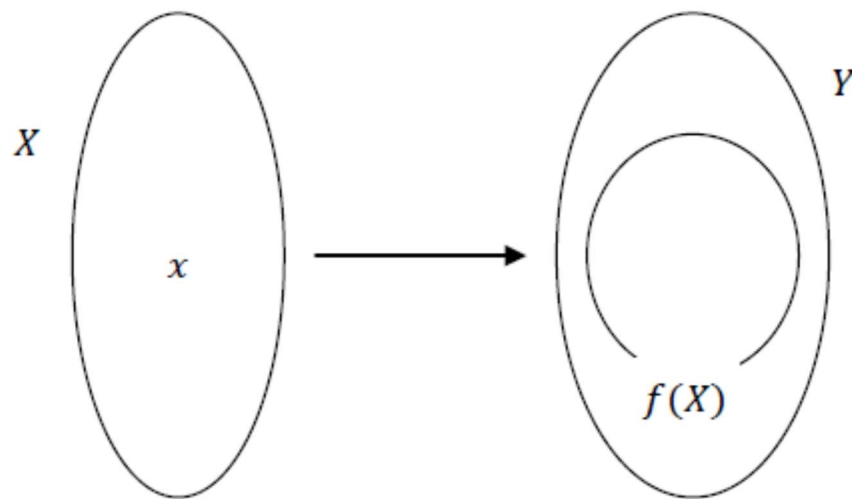
Notation:

Let f be a function from set X to set Y . Since for each $x \in X$ there exists a uniquely determined $y \in Y$ with $(x, y) \in f$, we write $y = f(x)$, and refer to $f(x)$ as the **image of x under f** .

We also write $f: X \rightarrow Y$ to indicate that f is a function which transforms or maps each element of X to a uniquely determined element of Y .

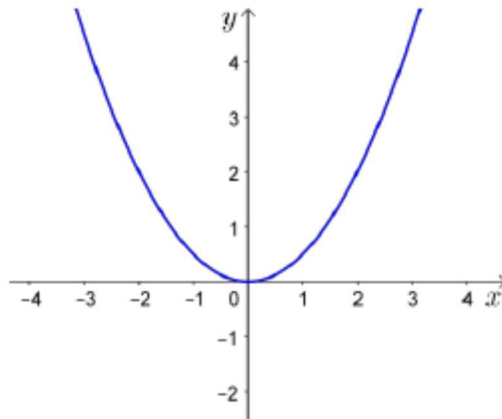
The set X is called the **domain** of f and the set Y is called the **codomain** of f .

The **range** of f is the set of images of all elements of X under f and it is denoted by $f(X)$. Hence, $f(X) = \{f(x): x \in X\}$ and it is a subset of the codomain ($f(X) \subseteq Y$).



When the sets X and Y are infinite sets, it is impossible to draw a digraph representation of a function $f: X \rightarrow Y$. We can use however, the more traditional idea of *graphing a function* to give a geometric interpretation of the function.

Consider for example the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$. The graph of f is given below:



The horizontal axis labelled x denotes the domain $\mathbb{R}$ of f and the vertical axis labelled y denotes the codomain $\mathbb{R}$ of f .

Note that the range of f is the set of non-negative real numbers.

The plane denotes the Cartesian product $\mathbb{R} \times \mathbb{R}$ and the curve consists of those points (x, y) in $\mathbb{R} \times \mathbb{R}$ for which $y = f(x)$.

Properties of Functions

Definition:

Let $f: A \rightarrow B$ be a function.

- We say that f is **injective** or **one-to-one** if

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2 \quad \forall a_1, a_2 \in A$$

This is logically equivalent to

$$a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)$$

An injective function never repeats values, Different inputs give different outputs.

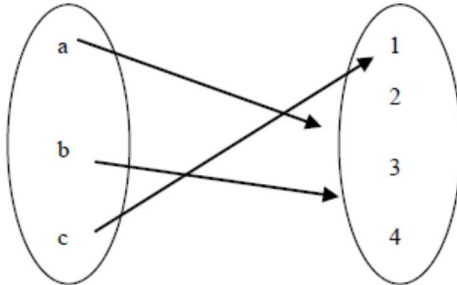
- We say that f is **surjective** or **onto** if the range of f coincides with its codomain. This means that for every $b \in B$ there exists (at least) one $a \in A$ with $f(a) = b$. Each element of the codomain is the image of some element of the domain.
- We say that f is **bijective** if f is both injective and surjective.

When dealing with functions between finite sets (and given the digraph representations of such functions), it is easy to determine if they are injective or surjective.

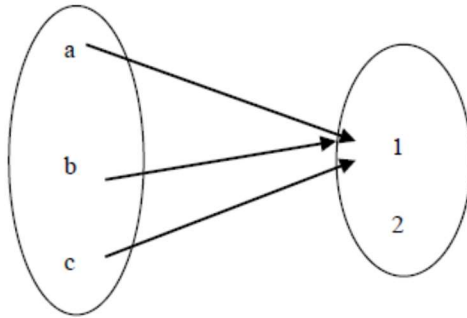
Example 3:

Determine whether each of the functions represented below is injective or surjective. Which of them are bijective?

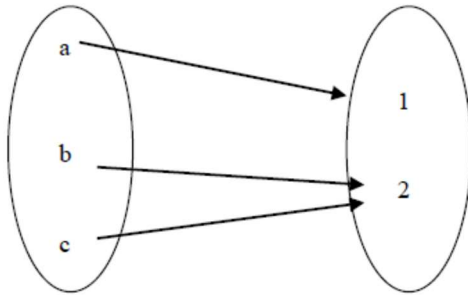
(a)



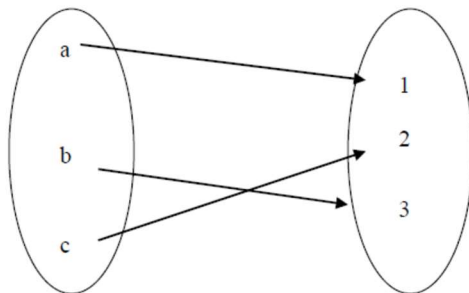
(b)



(c)



(d)



Solution:

- (a) The function is injective since each element of the domain is mapped to a distinct element of the codomain. It is not surjective since there is no element x in the domain of the function such that $f(x) = 2$. In other words, the element 2 is in the codomain of the function but not in its range, therefore codomain and range do not coincide. Since the function is not surjective it is not bijective.
- (b) The is not injective since $f(a) = f(b) = f(c) = 1$. It is not surjective since there is no element x in the domain of the function such that $f(x) = 2$. In other words, the element 2 is in the codomain of the function but not in its range, therefore codomain and range do not coincide. Since the function is neither injective nor surjective it is not bijective.
- (c) The function is not injective since $f(b) = f(c) = 2$. The function is surjective since the range and codomain are the same. Since the function is not injective, it is not bijective.
- (d) The function is injective since each element of the domain is mapped to a distinct element of the codomain. The function is surjective since the range and codomain are the same. Since the function is both injective and surjective, it is bijective.

It gets more complicated when we are dealing with functions between infinite sets.

In general, it is easier to show that a function is NOT injective or surjective.

Example 4:

Determine whether the function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = x^2$ is injective or surjective or both. Justify your answer.

Solution:

The function is not injective. To prove that, it is sufficient to find integers a_1 and a_2 with $a_1 \neq a_2$ and $f(a_1) = f(a_2)$. One such example is $f(-1) = 1 = f(1)$.

The function is not surjective. To prove this, it is sufficient to find an integer a in the codomain of f that is not in the range of f . Any negative integer would do. For example, $f(x) \neq -1$ for any value of $x \in \mathbb{Z}$.

Note:

If the domain of the function above was the set of positive real numbers, then the function would be injective.

If in addition the codomain was the set of square numbers, then the function would also be surjective.

Example 5:

Show that the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 4x + 3$ is bijective.

Solution:

Suppose that $a_1, a_2 \in \mathbb{R}$, with $f(a_1) = f(a_2)$. Then:

$$f(a_1) = f(a_2) \Rightarrow 4a_1 + 3 = 4a_2 + 3 \Rightarrow 4a_1 = 4a_2 \Rightarrow a_1 = a_2$$

This is true for any real numbers a_1, a_2 . Therefore, f is injective.

Let $b \in \mathbb{R}$. Can we find $a \in \mathbb{R}$ such that $f(a) = b$?

$$f(a) = b \Rightarrow 4a + 3 = b \Rightarrow 4a = b - 3 \Rightarrow a = \frac{b - 3}{4}$$

So, for any $b \in \mathbb{R}$, $f\left(\frac{b-3}{4}\right) = b$. Hence, f is surjective.

Since f is both injective and surjective, we conclude that f is bijective.

Example 6:

Determine whether the function $f: \mathbb{N} \rightarrow \mathbb{N}$ given by $f(x) = 2^x$ is injective or surjective or both. Find the range of f .

Solution:

The range of f is the set $\{2, 4, 8, 16, 32, \dots\}$. Since the range is not equal to the codomain, we conclude that f is not surjective.

Suppose that $a_1, a_2 \in \mathbb{R}$, with $f(a_1) = f(a_2)$. Then:

$$f(a_1) = f(a_2) \Rightarrow 2^{a_1} = 2^{a_2} \Rightarrow a_1 = a_2$$

This is true for any real numbers a_1, a_2 . Therefore, f is injective.

Example 7:

Determine whether the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} 2x - 3 & \text{if } x \geq 1 \\ x + 1 & \text{if } x < 1 \end{cases}$$

is injective, surjective or both.

Solution:

Note that $f(2) = 2 \cdot 2 - 3 = 1 = f(0) = 0 + 1$. Hence, f is not injective.

Let $b \in \mathbb{R}$.

If $b \geq -1$, then $2a - 3 = b \Rightarrow a = \frac{b+3}{2} \geq 1$ and $f\left(\frac{b+3}{2}\right) = b$.

If $b < -1$, then $a + 1 = b \Rightarrow a = b - 1 < -2 < 1$ and $f(b - 1) = b$.

Hence, f is bijective.

Inverse Function

Definition:

Any function $f: A \rightarrow B$ is a relation, so we can form the inverse relation f^{-1} .

If the inverse relation is itself a function, we say that f is **invertible** and we write: $f^{-1}: B \rightarrow A$ for the **inverse function**.

The function f consists of pairs of the form (a, b) , where $f(a) = b$. When f is invertible, the inverse function f^{-1} contains the pairs (b, a) where $a = f^{-1}(b)$. Hence, for an invertible function f , if $f(a) = b$, then

$$f^{-1}(b) = a = f^{-1}(f(a))$$

In other words, the inverse function reverses the effect of the original function.

From the functions in Example 6, only (d) is invertible. The inverse relations are given by simply reversing the arrows in the digraph. When doing so, (a) and (b) do not give functions, since no element will be associated with 2, and (c) does not give a function since two elements (b and c) will be associated with the element 2. Note, that the function in part (d) is the only one of these functions that is bijective.

We can use the same idea (of reversing the arrows) to determine the inverse of more complex invertible functions that are given algebraically.

For example, let us find the inverse of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 4x + 3$ from Example 8. We can break down the effect of f in the following steps:

- start with x
- multiply by 4 to obtain $4x$
- add 3 to obtain $4x + 3$

We reverse these steps as follows:

- start with y .
- subtract 3.
- divide by 4.

Hence, we can conclude that the inverse function of f is the function $f^{-1}: \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f^{-1}(x) = \frac{x - 3}{4}$$

We could obtain the same result algebraically by solving for x the following expression:

$$y = f(x) = 4x + 3 \Rightarrow y - 3 = 4x \Rightarrow x = \frac{y - 3}{4}$$

and interchanging x and y (since we usually use x to denote the input) to obtain:

$$f^{-1}(x) = \frac{x - 3}{4}$$

Note that both the functions above that are invertible are also bijective. This is not a coincidence.

Theorem:

A function $f: A \rightarrow B$ is invertible if and only if f is bijective.

Example 8:

Let $A = \{x: x \in \mathbb{R} \text{ and } x \neq 1\}$ and $f: A \rightarrow A$ be given by

$$f(x) = \frac{x}{x-1}$$

Determine if f is injective or surjective or both. If f is bijective determine the inverse function.

Solution:

First, we show that f is injective.

Suppose that $a_1, a_2 \in A$, with $f(a_1) = f(a_2)$. Then:

$$\begin{aligned} f(a_1) = f(a_2) &\Rightarrow \frac{a_1}{a_1-1} = \frac{a_2}{a_2-1} \Rightarrow a_1(a_2-1) = a_2(a_1-1) \\ &\Rightarrow a_1a_2 - a_1 = a_2a_1 - a_2 \Rightarrow a_1 = a_2 \end{aligned}$$

Hence, f is injective.

Let $b \in A$ (the codomain of f). Can we find $a \in A$ (the domain of f) such that $f(a) = b$?

$$\begin{aligned} f(a) = b &\Rightarrow \frac{a}{a-1} = b \Rightarrow a = ab - b \Rightarrow b = ab - a \\ &\Rightarrow b = a(b-1) \Rightarrow a = \frac{b}{b-1} \end{aligned}$$

Hence, f is surjective.

Since f is both injective and surjective, it is bijective, and hence, invertible. The inverse function satisfies $f^{-1}(b) = a$ whenever $f(a) = b$ but we showed above that

$$a = \frac{b}{b-1}$$

Therefore, the inverse function is given by $f^{-1}: A \rightarrow A$ where

$$f^{-1}(x) = \frac{x}{x-1}$$

Note: The function f is its own inverse.

Composition of Functions

We now look at the composition of functions, that will be a little easier to handle than the composition of relations seen earlier.

Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions. Then the composite relation $g \circ f$ between A and C consists of the pairs (a, c) where $(a, b) \in f$ and $(b, c) \in g$ for some $b \in B$. However, $b = f(a)$ is uniquely determined by a since f is a function and $c = g(b)$ is uniquely determined since g is a function. Hence, $c = g(f(a))$ is uniquely determined by a , and so the composition of f and g is also a function.

Therefore: $g \circ f: A \rightarrow C$ is the function given by $(g \circ f)(x) = g(f(x))$.

Example 9:

Consider the functions $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^2$ and $g(x) = 4x + 3$ respectively. Calculate the functions $g \circ f$, $f \circ g$, $f \circ f$, $g \circ g$, $g^{-1} \circ f$ and $g^{-1} \circ g$.

Solution:

$$(g \circ f)(x) = g(f(x)) = g(x^2) = 4x^2 + 3$$

$$(f \circ g)(x) = f(g(x)) = f(4x + 3) = (4x + 3)^2 = 16x^2 + 24x + 9$$

$$(f \circ f)(x) = f(f(x)) = f(x^2) = (x^2)^2 = x^4$$

$$(g \circ g)(x) = g(g(x)) = g(4x + 3) = 4(4x + 3) + 3 = 16x + 15$$

$$(g^{-1} \circ f)(x) = g^{-1}(f(x)) = g^{-1}(x^2) = \frac{x^2 - 3}{4}$$

$$(g^{-1} \circ g)(x) = x \quad (\text{by definition})$$

The pigeonhole principle

Let $f: A \rightarrow B$ be a function where A and B are finite sets. Suppose that A contains n elements ($|A| = n$) $a_1, a_2, \dots, a_n$. If $|A| > |B|$ then at least one value of f occurs more than once. In other words, $f(a_i) = f(a_j)$ for some i and j where $i \neq j$.

This is called the **pigeonhole principle**. If for example we have n pigeons and m pigeonholes, where $n > m$, and we want to put every pigeon in a pigeonhole, then at least two pigeons will be put in the same pigeonhole.

Some of the problems we will see can be solved easily with logical arguments. Some others will require the use of the pigeonhole principle. In the later cases, the most challenging part would be to come up with an appropriate function.

Example 10:

How many different surnames must appear in the telephone directory to guarantee that at least two of the surnames

- (a) begin with the same letter of the alphabet?
- (b) begin with the same letter and end with the same letter of the alphabet?

Solution:

- (a) Let A be the set of surnames and B be the set of letters of the English alphabet (26). Let $f: A \rightarrow B$ be a function that assigns each surname to its first letter. For example, $f(\text{Michaelides}) = M$. Since $|B| = 26$, the pigeonhole principle guarantees that two of the surnames will begin with the same letter provided that $|A| > 26$. Therefore, the telephone directory must contain at least 27 different surnames to guarantee that at least two of them begin with the same letter.
- (b) Let A be the set of surnames, B be the set of letters of the English alphabet and C be the Cartesian product $C = B \times B$ of pairs of letters of the alphabet. Let $f: A \rightarrow C$ be a function that assigns each surname to the pair of letters corresponding to its first and last letters. For example, $f(\text{Michaelides}) = (M, s)$. Note that $|C| = |B| \times |B| = 26 \times 26 = 676$. The pigeonhole principle guarantees that two of the surnames will begin and end with the same pair of letters provided that $|A| > 676$. Therefore, the telephone directory must contain at least 677 different surnames to guarantee that at least two of them begin and end with the same letters.

Example 11:

How many different numbers must be selected from the set

$S = \{1, 2, 3, 4, 5, 6, 7, 8\}$ so that at least two of the numbers selected sum to 9?

Solution:

The pairs of numbers that sum to 9 are (1,8), (2,7), (3,6) and (4,5). Let A be the set of numbers selected from S and let B be the set

$B = \{\{1, 8\}, \{2, 7\}, \{3, 6\}, \{4, 5\}\}$. Let $f: A \rightarrow B$ be a function that assigns the element of A to the pair of B it belongs. For example,

$$f(3) = \{3, 6\} = f(6)$$

By the pigeonhole principle, if $|A| > |B| = 4$ at least two of the numbers selected from S will be assigned to the same element of B (hence, they will sum to 9). Therefore, at least 5 numbers must be selected from S so that at least two of them sum to 9.

Example 12:

Prove that in any party of 2 or more people, there are at least 2 people with the same number of friends (in the party).

Solution:

Case 1: Everyone has at least one friend in the party. Let A be the set of people in the party and say $|A| = n$. Then each person in the party can have a minimum of 1 friend and a maximum of $n - 1$ friends. Let B be the set $B = \{1, 2, 3, \dots, n - 1\}$. Let $f: A \rightarrow B$ be a function that assigns to each person in the party the number of friends they have in the party. Since $|A| > |B|$ the pigeonhole principle guarantees that at least two people will be assigned to the same number, hence, at least two people in the party will have the same number of friends in the party.

Case 2: At least one person has no friends in the party. If there are 2 people with no friends in the party, then we are done since two people will have 0 friends. So, let us assume that there is only one person in the party with no friends. Let A be the set of the rest of the people in the party (that have at least one friend) and say $|A| = n - 1$. Then each of these people can have a minimum of 1 friend and a maximum of $n - 2$ friends. Let B be the set $B = \{1, 2, 3, \dots, n - 2\}$. Let $f: A \rightarrow B$ be a function that assigns to each person in A the number of friends they have in the party. Since $|A| > |B|$ the pigeonhole principle guarantees that at least two people will be assigned to the same number, hence, at least two people in the party will have the same number of friends in the party.

The pigeonhole principle can be extended as follows:

Consider the function $f: A \rightarrow B$ where A and B are finite sets with $|A| > k|B|$ for some natural number k . Then, there is at least one value of f that occurs at least $k + 1$ times.

This result holds since, if each value of f occurs at most k times, A would contain at most $k|B|$ elements.

Example 13:

How many different surnames must appear in the telephone directory to guarantee that at least 5 of the surnames begin with the same letter and end with the same letter of the alphabet?

Solution:

Let A be the set of surnames in the telephone directory, B be the set of letters of the English alphabet and C the Cartesian product $C = B \times B$ of pairs of letters of the alphabet. Let $f: A \rightarrow C$ be a function that assigns each surname to the pair of letters corresponding to its first and last letters. For example, $f(\text{Michaelides}) = (M, s)$. Note that

$$|C| = |B| \times |B| = 26 \times 26 = 676.$$

To have at least 5 ($k + 1 = 5$) surnames begin with the same letter and end with the same letter, we require that $|A| > 4|C| = 4 \times 676 = 2704$.

Therefore, the telephone directory must contain at least 2705 surnames to guarantee that at least 5 of the surnames begin with the same letter and end with the same letter of the alphabet.